# Carlota Lemos

Porto, Portugal | carlotamendeslemos@gmail.com | +351 967 009 311 | Personal Website
carlotamendeslemos | GitHub

## EDUCATION

**Sapienza Università di Roma** — *Rome, Italy*
*Bachelor's in Applied Computer Science and Artificial Intelligence* — *Sep 2026 – Jan 2027*

- Erasmus+ International Exchange Program

**Instituto Superior de Engenharia do Porto** — *Porto, Portugal*
*Bachelor's in Computer Engineering* — *Sep 2024 – Jul 2027*

- Current GPA: 14.8/20
- Completed Laboratory/Project II & III courses, receiving **two Certificates of Merit** for outstanding project performance; projects recognised among the top three out of 60+ projects developed during the respective semesters.

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, C/C++, JavaScript, TypeScript, Magik, Assembly

**Web Development & Frameworks:** React, Next.js, JavaFX, HTML5, CSS3, Tailwind CSS

**Databases & Storage:** Oracle SQL, PL/SQL

**Developer Tools & Environment:** Git, Docker, Jira, VS Code, IntelliJ IDEA, GNU Emacs, Jupyter Notebook

**Methodologies:** Agile (Scrum), Project Management Foundations, Business Analysis

## PROFESSIONAL EXPERIENCE

**Frontend Developer** — *Apr 2026 – Sep 2026*
*Adhara Academy* — *Remote*

- Responsible for developing the frontend core of a scalable medical question bank (QBank) using React and JavaScript.
- Implemented advanced question filters, dynamic exam simulators, and analytical dashboards for user metrics.
- Optimized application performance with mobile-first layouts, PWA offline support, and background preloading.

**Software Engineer Summer Intern** — *Jul 2026 – Aug 2026*
*CGI Portugal* — *Porto, Portugal*

- Completed technical training in GE Smallworld GIS and its native language, Magik, focusing on object-oriented design and asset data modeling for the E-REDES grid.
- Queried the VMDS database, executing spatial transactions, and writing Magik scripts using GNU Emacs.
- Analysed network topologies, vector geometries, and spatial contexts to understand infrastructure connectivity and identify data integrity issues.
- Completed the **Certified Associate in Project Management (CAPM)®** - Certification Path via CGI Academia, mastering foundational project management frameworks and agile methodologies.

## PROJECTS

**Malaria Detection in Blood Smear Images** — *Nov 2026 – Jan 2027*
*Academic Project*

- Developing a machine learning model that detects malaria-infected cells in microscopy images and tests how well it works on data from other labs.
- Comparing classical feature extraction with deep learning (pretrained CNNs) for image classification.
- Tools: Python, PyTorch, scikit-learn, OpenCV

**Air Traffic Control and Flight Management System** — *GitHub Repository*
*4th Semester University Integrative Project* — *Apr 2026 – Jun 2026*

- Co-developed a distributed air traffic control system, engineering flight simulation modules, a custom DSL compiler for syntax validation, and managing Scrum deliverables.
- Tools: Java, JavaFX, PL/SQL, C, ANTLR, Cisco Packet Tracer
- Grade: 19/20

**Portfolio Website (Frontend)** *GitHub Repository*
*Personal Project* *Feb 2026*

- Designed and developed a fully responsive personal portfolio website in Figma and React, featuring dynamic scroll animations and optimized assets.
- Tools: Next.js, TypeScript.

**Rail Freight Logistics Management System** *GitHub Repository*
*3rd Semester University Integrative Project* *Sep 2025 – Jan 2026*

- Co-developed a rail logistics system-engineering warehouse protocols, route scheduling, and optimized databases—while managing sprint planning and requirements elicitation.
- Tools: Java, JavaFX, PL/SQL, C, Assembly
- Grade: 18/20

**Railway Network Simulator** *GitHub Repository*
*2nd Semester University Integrative Project* *Feb 2025 – Jun 2025*

- Co-developed an interactive railway simulator, implementing network management, line tracking, and strategic optimization algorithms.
- Tools: Java, JavaFX, GraphStream, JUnit
- Grade: 18/20

**Image Reconstruction and Identification Using Eigenfaces** *GitHub Repository*
*1st Semester University Integrative Project* *Dec 2024 – Jan 2025*

- Co-developed a Java & Python facial recognition app, implementing Eigenfaces (PCA) for dimensionality reduction, automated face generation, and Euclidean distance pipelines.
- Tools: Java, Python, JUnit, la4j, CLI
- Grade: 20/20

## EXTRACURRICULAR EXPERIENCE

**Vice-President** *Jun 2026 – Present*
*Núcleo de Estudantes de Informática do ISEP* *Porto, Portugal*

- Lead and coordinate a 65-member student organization, overseeing multiple departmental operations to ensure strategic alignment and productivity.
- Manage internal workflows, member performance, and cross-departmental collaboration to optimize day-to-day operations and organizational growth.
- Oversee the planning, budgeting, and execution of large-scale academic and tech events, ensuring high-quality delivery and maximum student engagement.

**Ambassador** *Aug 2025 – Present*
*CGI Portugal Ambassadors Program* *Porto, Portugal*

- Represent CGI Portugal at university tech events, bridging the gap between students and the company's recruitment teams.

**Co-coordinator of the Design Department** *Jan 2026 – Jun 2026*
*Núcleo de Estudantes de Informática do ISEP* *Porto, Portugal*

- Co-coordinated a 14-member design team, managing internal workflows and ensuring consistent branding across all web platforms and social media.

**Member of the Design Department** *Jul 2025 – Jan 2026*
*Núcleo de Estudantes de Informática do ISEP* *Porto, Portugal*

- Designed UI/UX visual assets and developed event websites using Figma.

**Member of the Informatics Department** *Oct 2024 – Jul 2025*
*Núcleo de Estudantes de Informática do ISEP* *Porto, Portugal*

- Developed and maintained the official student association website using React and Tailwind CSS.

## LANGUAGES

**Portuguese:** Native Proficiency.

**English:** Advanced Proficiency - B2 Level.